

LINUX HACKATHON

TASK 1:

Set up a Nginx server to serve the custom HTML page provided by the organizers.

```
root@EXPLORER:~#22 sudo apt install nginx -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
nginx is already the newest version (1.24.0-2ubuntu7.18).
0 upgraded, 0 newly installed, 0 to remove and 137 not upgraded.
root@EXPLORER:~#23 systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset>
   Active: active (running) since Thu 2026-10-08 03:48:30 UTC; 13min ago
     Docs: man:nginx(8)
   Process: 996 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_pr>
   Process: 997 ExecStart=/usr/sbin/nginx -g daemon on; master_process on;>
  Main PID: 1036 (nginx)
    Tasks: 13 (Limit: 9303)
   Memory: 9.1M (peak: 20.2M)
      CPU: 85ms
   CGroup: /system.slice/nginx.service
           └─1036 "nginx: master process /usr/sbin/nginx -g daemon on; ma>
             └─1038 "nginx: worker process"
               └─1039 "nginx: worker process"
                 └─1040 "nginx: worker process"
                   └─1041 "nginx: worker process"
                     └─1042 "nginx: worker process"
                       └─1043 "nginx: worker process"
                         └─1044 "nginx: worker process"
                           └─1045 "nginx: worker process"
                             └─1046 "nginx: worker process"
                               └─1047 "nginx: worker process"
                                 └─1048 "nginx: worker process"
                                   └─1049 "nginx: worker process"

Oct 08 03:48:30 EXPLORER systemd[1]: Starting nginx.service - A high perfor>
Oct 08 03:48:30 EXPLORER systemd[1]: Started nginx.service - A high perform>
```

```
root@EXPLORER:~#24 curl -I http://10.10.110.79:3923/test/index.html
HTTP/1.1 200 OK
Content-Length: 382
Connection: Keep-Alive
Date: Thu, 08 Oct 2026 04:02:29 GMT
Cache-Control: no-cache
Last-Modified: Wed, 07 Oct 2026 14:46:01 GMT
X-Lastmod3: 1791384361000
Accept-Ranges: bytes
Content-Type: text/html; charset=utf-8
X-Content-Type-Options: nosniff
Vary: Origin, PW, Cookie

root@EXPLORER:~#25 wget "http://10.10.110.79:3923/test/index.html" -O custom.html
--2026-10-08 04:04:04-- http://10.10.110.79:3923/test/index.html
Connecting to 10.10.110.79:3923... connected.
HTTP request sent, awaiting response... 200 OK
Length: 382 [text/html]
Saving to: 'custom.html'

custom.html      100%[=====]      382  --.-KB/s   in 0.001s

2026-10-08 04:04:04 (468 KB/s) - 'custom.html' saved [382/382]
```

```

root@EXPLORER:~#27 cp custom.html /var/www/html/index.html
root@EXPLORER:~#28 curl http://localhost
<!DOCTYPE html>
<html>
<head>
  <title>Linux Community - OpenHack Hackathon</title>
</head>
<body>
  <h1>Linux Community - OpenHack Hackathon</h1>

  <p>Conducted on October 8th, 2026!</p>

  <p>Name: [Your Name]</p>
  <p>Roll No: [Your Roll No]</p>

  <p>Good luck!</p>

  <p><i>Only add your Name and Roll no when you do task 3</i></p>
</body>
</html>root@EXPLORER:~#29 mkdir -p ~/nginx-hackathon/task1n/task1

```

Commands:

sudo apt update

sudo apt install nginx -y

sudo systemctl start nginx

curl -I http://10.10.110.79:3923/test/index.html

wget "http://10.10.110.79:3923/test/index.html" -O custom.html

cp custom.html /var/www/html/index.html

TASK 2:

Access the hosted HTML page from another computer on the same network.

Linux Community - OpenHack Hackathon

Conducted on October 8th, 2026!

Name: [Your Name]

Roll No: [Your Roll No]

Good luck!

Only add your Name and Roll no when you do task 3

Commands:

hostname -I

curl <http://172.19.123.521>

TASK 3:

From another computer, modify the custom HTML page and verify that the changes are reflected on the hosted page.

```
Command not found, but there are 24 similar ones.  
root@EXPLORER:~69 nano /var/www/html/index.html  
root@EXPLORER:~70 curl http://localhost  
<!DOCTYPE html>  
<html>  
<head>  
  <title>Linux Community - OpenHack Hackathon</title>  
</head>  
<body>  
  <h1>Modified html</h1>  
</body>  
</html>  
root@EXPLORER:~71
```

Modified html

Commands:

sudo apt install openssh-server -y

sudo systemctl start ssh

whoami

hostname -I

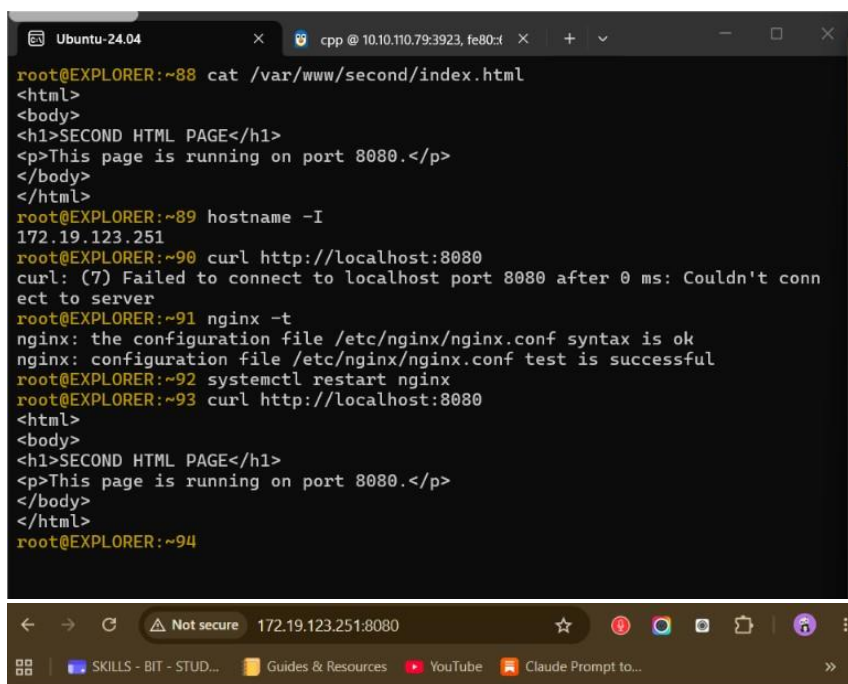
ssh [root@172.19.123.521](ssh:root@172.19.123.521)

nano /var/www/html/index.html

curl <http://172.19.123.521>

TASK 4:

Run a different HTML page on a separate port and verify that it can be accessed.



The image shows a terminal window and a web browser. The terminal window displays the following commands and output:

```
root@EXPLORER:~88 cat /var/www/second/index.html
<html>
<body>
<h1>SECOND HTML PAGE</h1>
<p>This page is running on port 8080.</p>
</body>
</html>
root@EXPLORER:~89 hostname -I
172.19.123.251
root@EXPLORER:~90 curl http://localhost:8080
curl: (7) Failed to connect to localhost port 8080 after 0 ms: Couldn't connect to server
root@EXPLORER:~91 nginx -t
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
root@EXPLORER:~92 systemctl restart nginx
root@EXPLORER:~93 curl http://localhost:8080
<html>
<body>
<h1>SECOND HTML PAGE</h1>
<p>This page is running on port 8080.</p>
</body>
</html>
root@EXPLORER:~94
```

The web browser shows the URL `172.19.123.251:8080` and displays the content of the second HTML page.

SECOND HTML PAGE

This page is running on port 8080.

Commands:

```
mkdir -p /var/www/second
```

```
nano /var/www/second/index.html
```

```
nano /etc/nginx/sites-available/second
```

```
ln -s /etc/nginx/sites-available/second /etc/nginx/sites-enabled/second
```

```
nginx -t
```

```
systemctl restart nginx
```

<http://172.19.123.521:8080>

Sites Referred :

<https://dev.to/anneusang/setting-up-nginx-to-serve-a-simple-html-page-2p4k>

<https://www.digitalocean.com/community/tutorials/how-to-install-nginx-on-ubuntu-20-04>

<https://jay75chauhan.medium.com/step-by-step-guide-to-hosting-a-website-with-nginx-on-linux-7d24e1857db4>

<https://unix.stackexchange.com/questions/150523/bash-iptables-command-not-found>